

CSC4026Z Chat Server Documentation

User Messages

These messages deal with Users. The definitions below use [these type indicators](#).

SET_USERNAME

Use this to change your username. Usernames **MUST** be unique, you can't pick the username of another active session. The exclusion to this rule is that a Wireguard connection **MAY** reclaim a username from it's previous session. Let's say a connection dies because the user's chat program exited. The server session will be cleared in 60s. Before the 60s are up:

- [Cleartext users](#) **CANNOT** reclaim usernames, they must wait for old sessions to expire
- [Wireguard users and Extended Wireguard users](#) **MAY** immediately set their username to the old username before the old session has expired

Additionally, [Cleartext users](#) **MUST** begin their usernames with the string 'clear-'. A valid username is 'clear-fancybear', but 'fancybear' is *not* a valid username for a [Cleartext users](#). Usernames **CANNOT** contain the character ':'. Usernames are UTF-8 encoded.

This message **MAY** arrive unsolicited. If you are joined to a channel where another user changes their username, then you will receive this message to inform you of the change. You will only receive one notification per change. In those cases,

`response_handle` will be `None`.

Request

```
SET_USERNAME_request {  
    u8      request_type = 13  
    u32     session  
    u32     request_handle  
    s[20]   username      # New username  
}
```

Response

```
SET_USERNAME_response {
    u8      response_type = 34
    u32      session
    u32      response_handle    # Only filled in for the sender
    s[20]    old_username      # Previous username
    s[20]    new_username      # New username
}
```

USER_LIST

Use this to fetch a list of all current users. A full user list won't always fit in a single UDP packet, so the message allows for rudimentary pagination. The server will return a list of up to 20 usernames. If there are more than 20 users on the server, then the `next_page` field will be `True`. You **MAY** repeat the list request, and include the `offset = 10`, to skip the first 10 users, `offset = 20` to skip the first 20 user, and so on.

Note that the server's user list **MAY** change between your list requests; if a user disconnected then your view of the user list will not be accurate. *shrug*

You **MAY** include a `channel` field, to request the list of users in a particular channel. If it's not present, then the entire user list is processed.

Request

```
USER_LIST_request {
    u8      request_type = 14
    u32      session
    u32      request_handle
    s[20]    channel        # Optional name of a channel to filter th
    u16      offset         # Leave out unless you are paging through
}
```

Response

```
USER_LIST_response {
    u8      response_type = 35
    u32      session
    u32      response_handle
    [s[20]]  users          # List of users
    bool     next_page      # Flag indicating whether more users
}
```

USER_MESSAGE

Use this to send a direct message to another user. Only that

user will see it.

Cleartext users **CANNOT** send DMs to Wireguard users or Extended Wireguard users, the server will reject them.

DMs are are UTF-8 encoded.

Request

```
USER_MESSAGE_request {
    u8      request_type = 12
    u32     session
    u32     request_handle
    s[20]   to_username      # Name of channel to send message to
    s[500]  message          # Message to send to channel
}
```

Response

There are two two responses to a *USER_MESSAGE*.

Sender Response

Senders of DMs **WILL** receive an OK response indicating the server processed the request.

Recipient Response

Recipients of DMS **WILL** receive this response message unsolicited.

```
USER_MESSAGE_response {
    u8      response_type = 33
    u32     session
    u32     response_handle = None
    s[20]   from_username    # Name of sending user
    s[500]  message          # Message sent to the user
}
```

WHOAMI

Use this to find out your current username. It is also provided in the CONNECT response, but maybe your client forgot?

Request

```
CHANNEL_INFO_request {
    u8      request_type = 11
    u32     session
    u32     request_handle
}
```

Response

```
CHANNEL_INFO_response {
  u8      response_type = 32
  u32      session
  u32      response_handle
  s[20]    username      # Your username
}
```

WHOIS

Use this to query information about a user.

Request

```
WHOIS_request {
  u8      request_type = 10
  u32      session
  u32      request_handle
  s[20]    username      # Name of user to query
}
```

Response

```
WHOIS_response {
  u8      response_type = 31
  u32      session
  u32      response_handle
  s[20]    username      # Username being queried
  [s[20]]  channels      # List of channel the user belongs to
  s[18]    transport     # One of ['wireguard', 'wireguard-clear']
  s[44]    wireguard_public_key # For Wireguard or Extended Wireguard
                                     # empty for cleartext users
}
```